

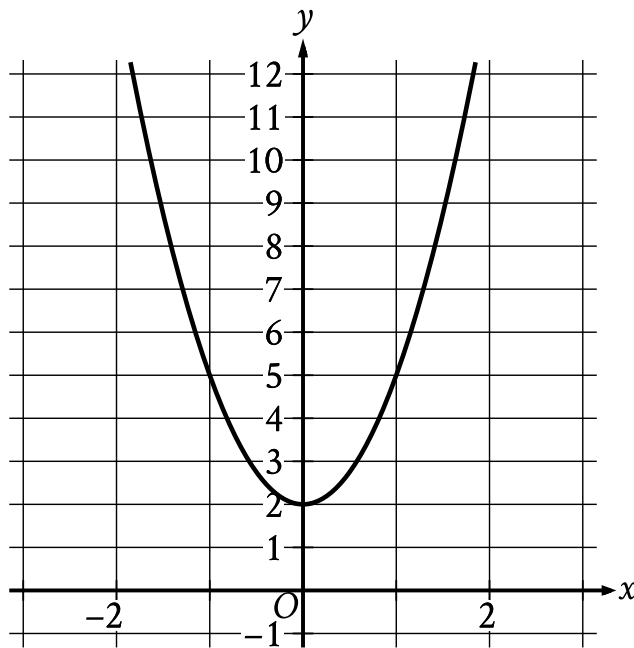
● EASY

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

04



- The parabola opens upward.
- The parabola passes through the following points:
 - (-1, 5)
 - (0, 2)
 - (1, 5)

The graph of the quadratic function $y = f(x)$ is shown. What is the vertex of the graph?

- A. 0, -2
- B. 0, -3
- C. 0, 2
- D. 0, 3

Q2

ADVANCED MATH

The function f is defined by $f(x) = 4 + \sqrt{x}$. What is the value of $f(144)$?

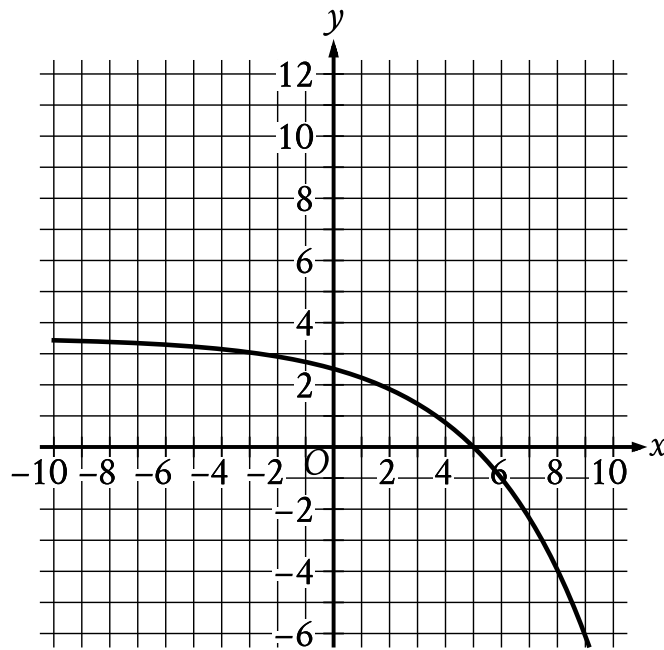
- A. 0
- B. 16
- C. 40
- D. 76

Q3

ADVANCED MATH

The function f is defined by $f(x) = x^3 + 15$. What is the value of $f(2)$?

- A. 20
- B. 21
- C. 23
- D. 24



- Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1 to quadrant 4.
 - In quadrant 2, the curve trends down gradually to the approximate point (0 comma 2.5).
 - In quadrant 1, the curve trends down sharply to point (5 comma 0).
 - In quadrant 4, the curve trends down sharply.
- The curve passes through the following points:
 - approximately (0 comma 2.5)
 - approximately (4 comma 0.8)

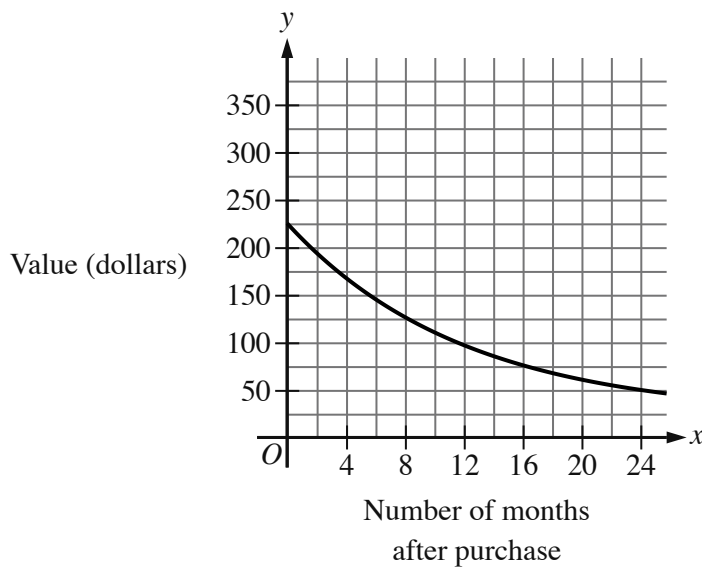
What is the x -intercept of the graph shown?

A. -5,0

B. 5,0

C. $-2,0$

D. $2,0$



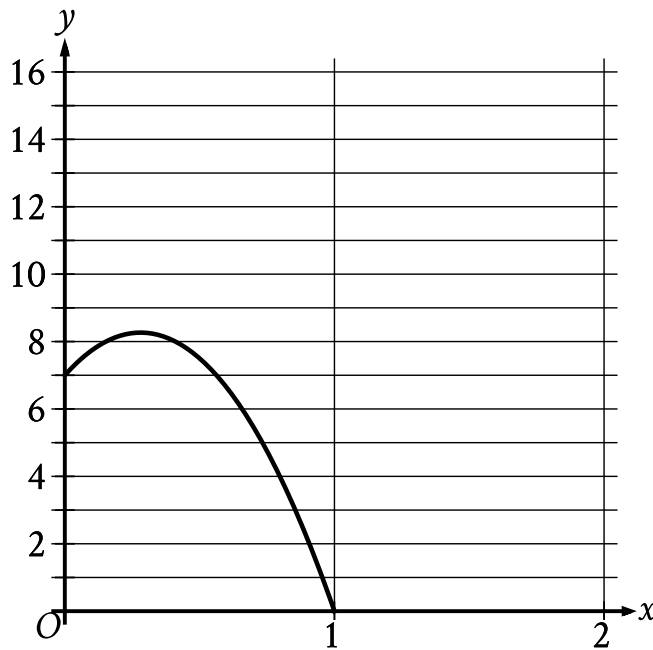
- Moving from left to right:
 - The curve is in quadrant 1.
 - The curve trends down gradually.
- The curve passes through the following approximate points:
 - (0 comma 225)
 - (24 comma 50)

The graph shown gives the estimated value, in dollars, of a tablet as a function of the number of months since it was purchased. What is the best interpretation of the y -intercept of the graph in this context?

- A. The estimated value of the tablet was \$ 225 when it was purchased.
- B. The estimated value of the tablet 24 months after it was purchased was \$ 225.
- C. The estimated value of the tablet had decreased by \$ 225 in the 24 months after it was purchased.

- D. The estimated value of the tablet decreased by approximately 2.25% each year after it was purchased.

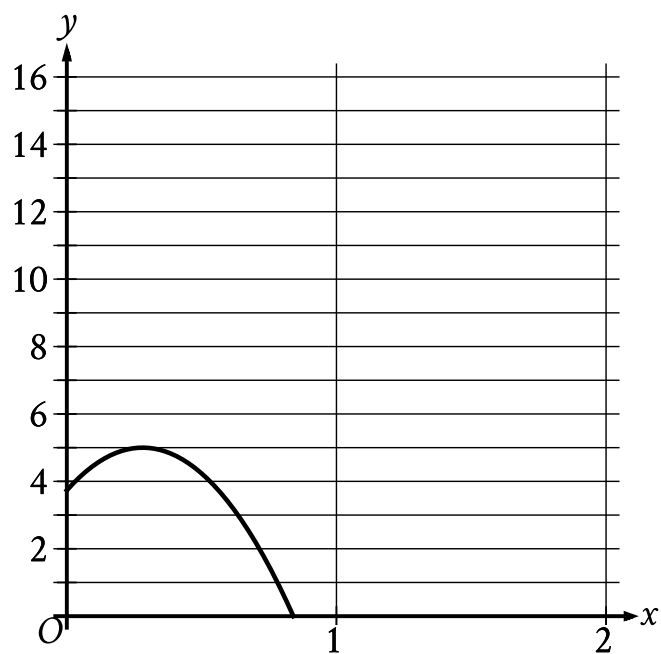
During the first part of an experiment, a ball was launched from a 7-foot-tall platform. The graph shows the height y , in feet, of the ball x seconds after it was launched during the first part of the experiment.



- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 8.3).
- The parabola passes through the following points:
 - (0 comma 7)
 - approximately (0.3 comma 8.3)
 - approximately (0.6 comma 7)

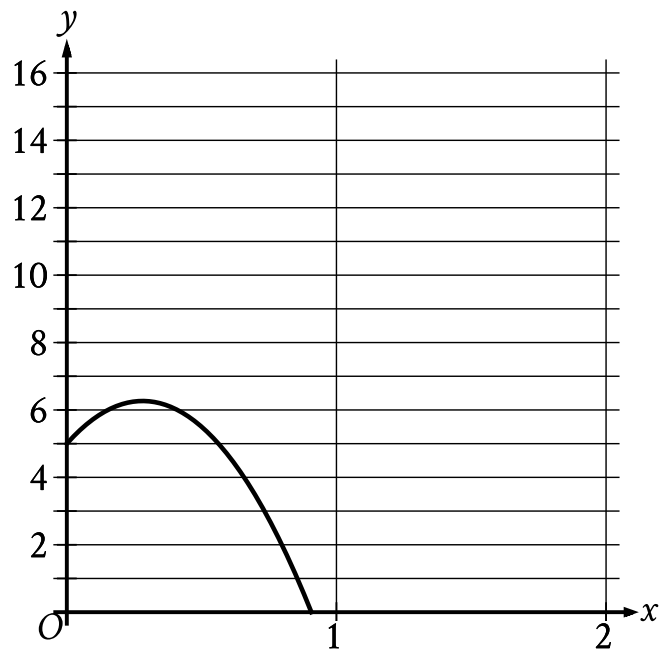
During the second part of the experiment, the ball was launched the same way, but from a platform that is 2 feet shorter than the first platform. Which of the following graphs could represent the height y , in feet, of the ball x seconds after it was launched during the second part of the experiment?

A.



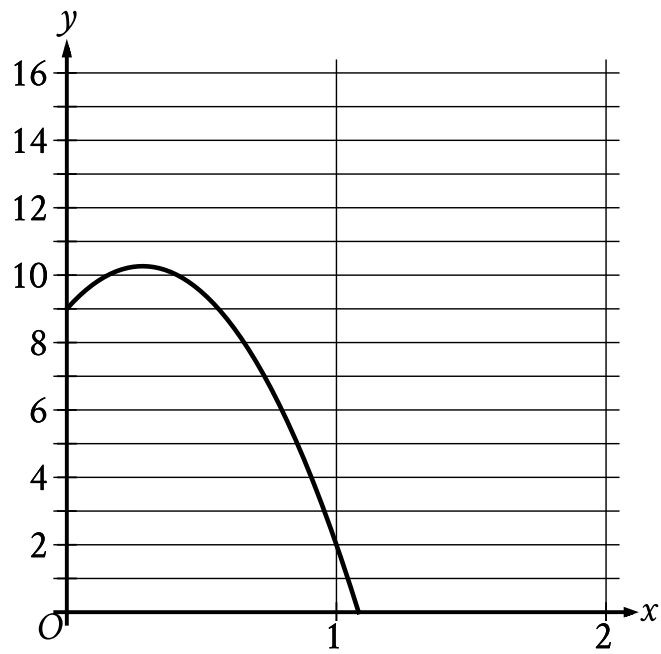
- The parabola opens downward.
- The vertex is at the approximate point (0.3 comma 5.0).
- The parabola passes through the following points:
 - approximately (0 comma 3.7)
 - approximately (0.3 comma 5.0)
 - approximately (0.6 comma 3.7)

B.



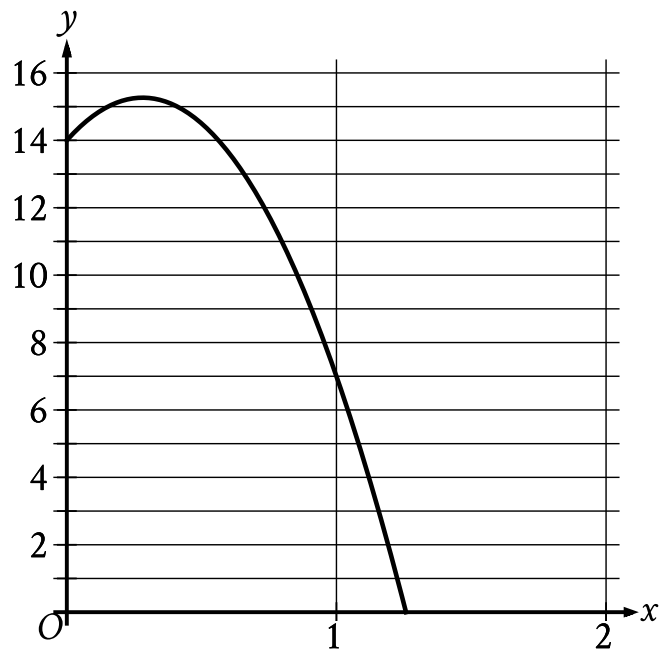
- The parabola opens downward.
- The vertex is at the approximate point (0.3, 6.3).
- The parabola passes through the following points:
 - (0, 5)
 - approximately (0.3, 6.3)
 - approximately (0.6, 5)

C.



- The parabola opens downward.
- The vertex is at the approximate point (0.3, 10.3).
- The parabola passes through the following points:
 - (0, 9)
 - approximately (0.3, 10.3)
 - approximately (0.6, 0)

D.



- The parabola opens downward.
- The vertex is at the approximate point (0.3, 15.3).
- The parabola passes through the following points:
 - (0, 14)
 - approximately (0.3, 15.3)
 - approximately (0.6, 14)

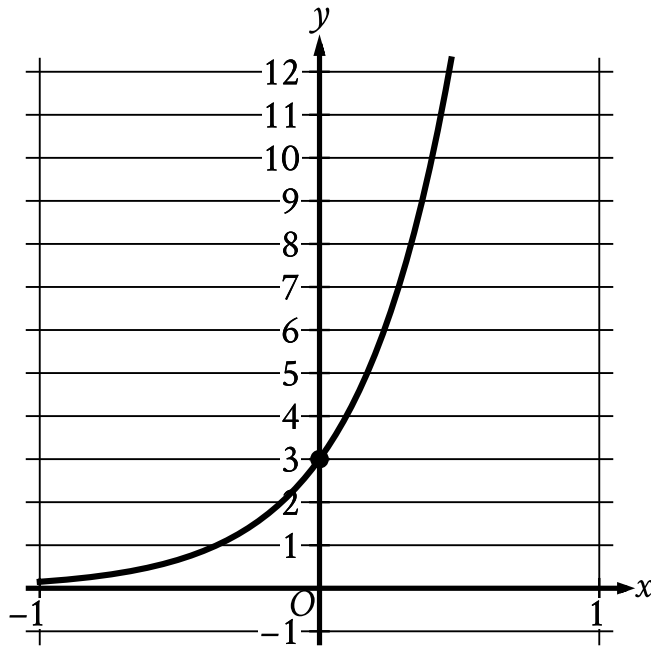
The function f is defined by $fx = \frac{1}{6x}$. What is the value of fx when $x = 3$?

A. $\frac{1}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{9}$

D. $\frac{1}{18}$



- Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2, the curve trends up sharply to the point $(0, 3)$.
 - In quadrant 1, the curve trends up sharply.
- The curve passes through the following points:
 - approximately $(-1, 0.2)$
 - $(0, 3)$

The graph of the exponential function f is shown, where $y = f(x)$. The y -intercept of the graph is $(0, y)$. What is the value of y ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN

ADVANCED MATH

The function h is defined by $hx = \frac{8}{5x+6}$. What is the value of $h2$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A ball is dropped from an initial height of 22 feet and bounces off the ground repeatedly. The function h estimates that the maximum height reached after each time the ball hits the ground is 85% of the maximum height reached after the previous time the ball hit the ground. Which equation defines h , where $h(n)$ is the estimated maximum height of the ball after it has hit the ground n times and n is a whole number greater than 1 and less than 10?

A. $h(n) = 22(0.22)^n$

B. $h(n) = 22(0.85)^n$

C. $h(n) = 850.22^n$

D. $h(n) = 85(0.85)^n$